

Symbols Used

Symbol	Meaning
ES	Early Start
EF	Early Finish
LS	Late Start
LF	Late Finish
d	Duration of activity
Lag	Lag time (can be +ve or -ve)

Total Float (TF) Formulas

Total Float = Maximum delay allowed without delaying project completion

Relationship	Total Float Formula
FS (Finish → Start)	$TF = LS - ES = LF - EF$
SS (Start → Start)	$TF = LS - ES$
FF (Finish → Finish)	$TF = LF - EF$
SF (Start → Finish)	$TF = LF - ES - d$

Free Float (FF) Formulas

Free Float = Delay allowed without delaying successor's early dates

Relationship	Free Float Formula
FS (Finish → Start)	$FF = ES(\text{successor}) - EF(\text{predecessor}) - \text{Lag}$

Relationship	Free Float Formula
SS (Start → Start)	$FF = ES(\text{successor}) - ES(\text{predecessor}) - \text{Lag}$
FF (Finish → Finish)	$FF = EF(\text{successor}) - EF(\text{predecessor}) - \text{Lag}$
SF (Start → Finish)	$FF = EF(\text{successor}) - ES(\text{predecessor}) - \text{Lag} - d$

Quick Comparison (Exam Tip Table)

Relationship	Float Depends On
FS	Finish of predecessor
SS	Start of predecessor
FF	Finish of both activities
SF	Start of predecessor & finish of successor

Key Exam Notes (Important)

- **Total Float can be zero or negative** (critical or delayed activities)
- **Free Float is always $\leq$ Total Float**
- In **FS**, Free Float is most commonly tested
- **SF is rare but tricky** → students often forget duration adjustment

If lag is already incorporated in ES / EF / LS / LF calculations,
then you must NOT subtract lag again in float formulas.

General Principle

- Lag considered during network scheduling (forward/backward pass)
→ Use simple float formulas
- Lag NOT yet applied
→ Explicitly subtract/add lag in formulas

Float Formulas WHEN LAG IS ALREADY CONSIDERED

♦ Total Float (TF)

Relationship	Total Float Formula
FS	$TF = LS - ES = LF - EF$
SS	$TF = LS - ES$
FF	$TF = LF - EF$
SF	$TF = LF - ES - d$

✓ No lag term anywhere

♦ Free Float (FF)

Relationship	Free Float Formula
FS	$FF = ES(\text{successor}) - EF(\text{predecessor})$
SS	$FF = ES(\text{successor}) - ES(\text{predecessor})$
FF	$FF = EF(\text{successor}) - EF(\text{predecessor})$
SF	$FF = EF(\text{successor}) - ES(\text{predecessor}) - d$

✓ Lag already embedded → **do not subtract lag**

When DO You Subtract Lag?

Situation	Action
Lag shown on arrow but not used in ES/EF	Subtract lag
Lag used in forward pass	Do NOT subtract lag
Software output (Primavera / MS Project)	Do NOT subtract lag
Hand-drawn network with raw dates	Subtract lag

One-Line Exam Rule (MEMORIZE)

“Lag is subtracted only once — either during scheduling OR during float calculation, never both.”

Typical Student Mistake

Lag considered in ES/EF

Lag subtracted again in FF formula

Double counting lag → wrong float
